CONVOLUTIONAL NEURAL NETWORK

INTRODUCTION

- CNN is a Deep Learning model mainly used for image processing and pattern recognition.
- CNNs consist of multiple layers, like the input layer, the Convolutional layer, the pooling layer, and the fully connected layers.

CONVOLUTION

- CNNs incorporate specialized layers, often multiple, designed for feature extraction.
- Within these layers are filters (also known as kernels).
- Filters perform pattern recognition, which is a key strength of CNNs.

FILTERS

- A filter is a small matrix (e.g., 3x3) that defines a specific pattern to search for in the input data.
- The filter slides (convolves) across the input data (e.g., an image), performing element-wise multiplication and summation at each location.
- The result of this convolution is a feature map, which indicates the presence and strength of the detected pattern in different regions of the input.

PATTERN MATCHING

1. Pattern matching in Convolutional Neural Networks (CNNs) means the process of detecting specific visual or spatial patterns (like edges, corners, or textures) in an input image by sliding a kernel (filter) over it.
2. Each kernel acts like a pattern detector; it looks for a certain feature and gives a high output when it matches that feature in the image.

WORKING

- Step 1 — Input: Start with a small region of an image (for example, a 6×6 matrix of pixel values).
- Step 2 — Kernel / Filter: A small matrix (like 3×3) called a kernel moves over the image.

Each kernel detects a specific type of pattern:

- Vertical edges (using Sobel vertical filter)
- Horizontal edges (using Sobel horizontal filter)
- Corners, curves, etc.
- Step 3 — Convolution

For each position:

- Multiply input region and kernel values elementwise.
- Add all products → gives one output value.
- Slide the kernel by 1 step (stride = 1) and repeat.

This forms an output feature map (like 4×4 if input is 6×6).

WEIGHT SHARING

- Weight sharing means that the same filter (set of weights) is used across the entire input image during convolution.
- Reason: Features like edges or shapes can appear anywhere in an image; top, bottom, left, or right.
- So, if one filter can detect a vertical edge, we want it to detect that edge no matter where it appears.

CNN ON COLOR IMAGES

- A color image has 3 channels:
  - 🔴 Red (R)
  - 🟢 Green (G)
  - 🔵 Blue (B)
- Each pixel is represented by three values (one for each color).
- A color image is a 3D matrix.
- Example:

If your image size = 6×6

Color → shape = (6, 6, 3)

RGB IMAGE REPRESENTATION

- An RGB image is a three-dimensional tensor object.
- The dimensions represent height, width, and the three color channels: Red, Green, and Blue (RGB).
- Example: An image of shape 6×6×3 has a height of 6, a width of 6, and 3 color channels.

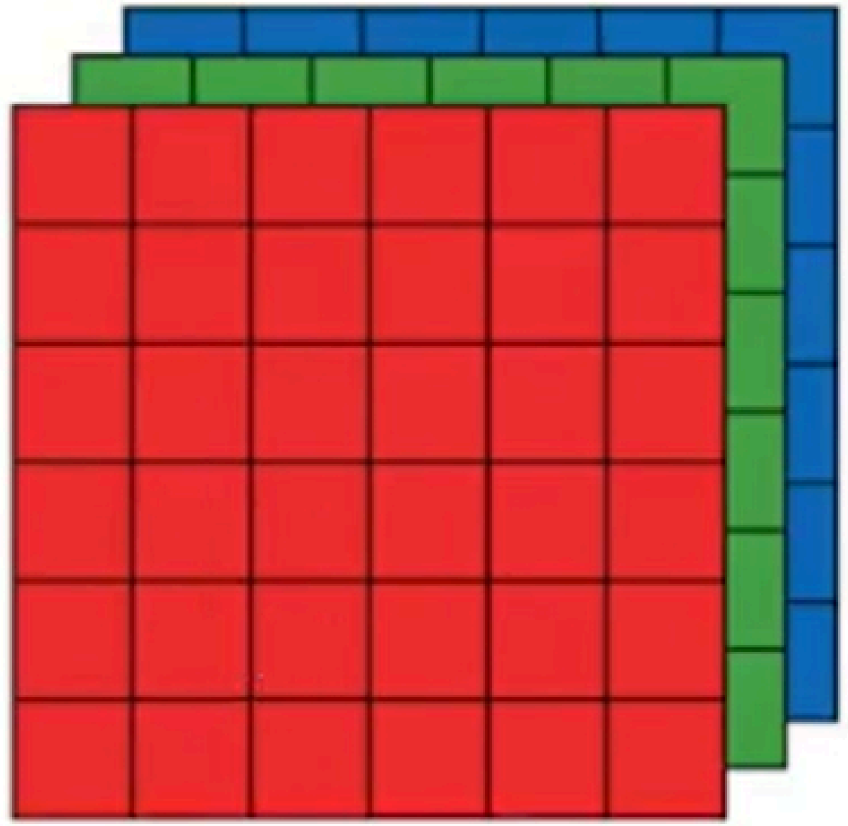
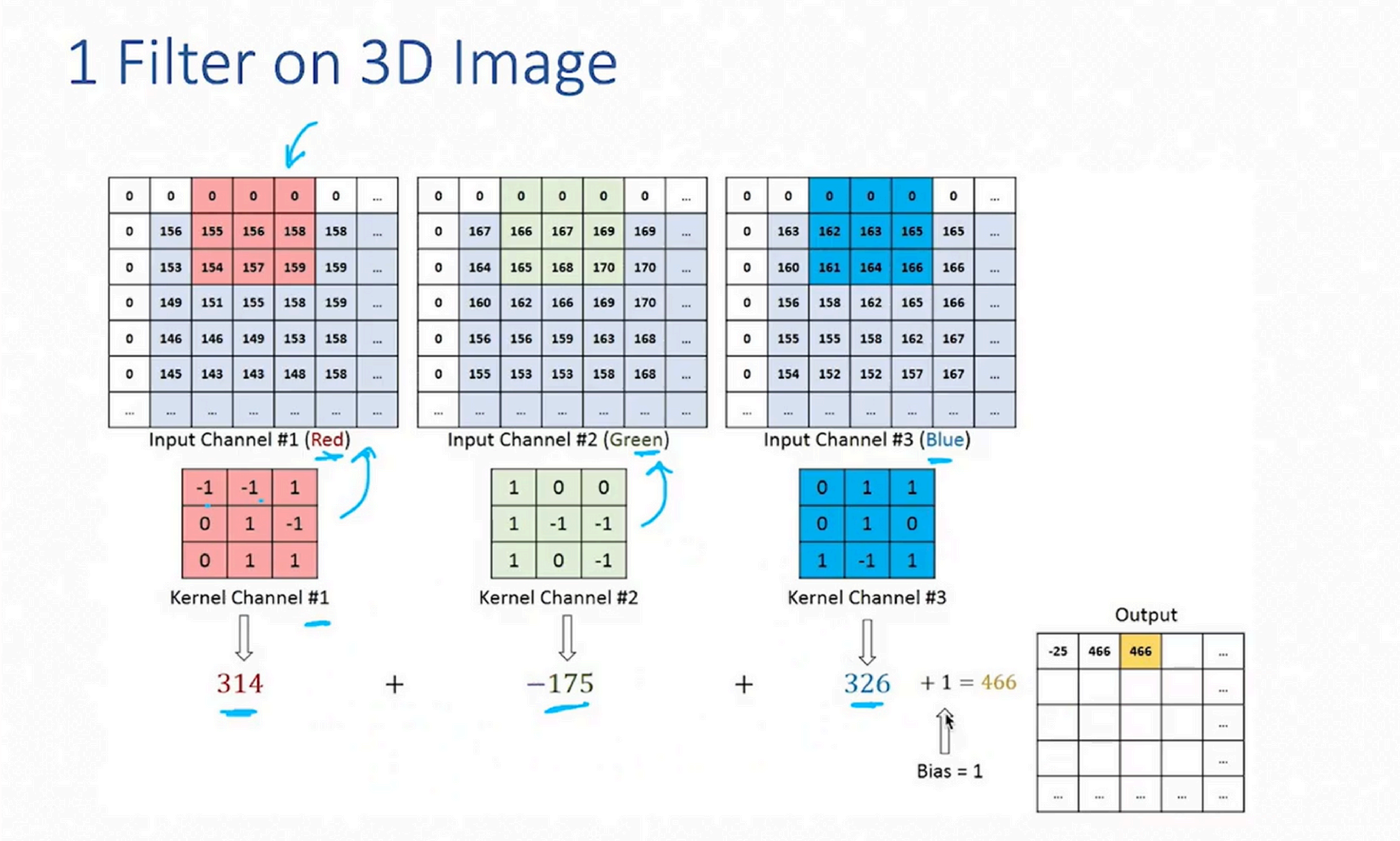


Image data
(6, 6, 3)

CONVOLUTION OPERATION ON COLOR IMAGES



CONVOLUTION OPERATION ON COLOR IMAGES

- Without bias:

  The output depends only on the exact weighted sum.
- If all values are 0, the output is 0.
- With bias:

  You can shift the output — for example, from 0 to 1 or from -1 to 0 — to make the network more flexible.
- Bias in CNN is a constant value added to the result of convolution.
- It helps shift the activation function output and allows the model to learn better even when all inputs are zero.

THANK YOU